What the Hadamard conjecture promises

Perfect balance, the orders we can reach, and why balance does not guarantee better rounding

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Version 3, prepared 22 October 2026, Malaysia time. AI assistance was used in the writing, derivations and verification code. Version 3 adds an account of the conjecture's history, constructions, computational frontier and asymptotic results. It merges the all-Hadamard addendum into the main text and shortens the application. The numerical study is unchanged from version 2.

A question about perfect balance

Imagine changing coordinates so that every original coordinate contributes equally in magnitude to every new coordinate. We also want the change to preserve lengths and angles. For real coordinates, these requirements force a remarkable object: a Hadamard matrix after normalization.

The Hadamard conjecture asks whether such perfect balance is possible in every dimension divisible by four. It does not ask whether orthogonal transformations exist. They exist in every dimension. It asks whether one can choose a particularly restrictive kind whose entries all have the same magnitude. The question has been open for more than ninety years. A Hadamard matrix is now known at every multiple of four below 2000, the last twelve announced in 2026, but no method is known that covers all orders. It is listed as JSP-000013 by the Justin Sun Prize. [1]

This article has three parts. The first explains what a Hadamard matrix is, why its order must be one, two or a multiple of four, and what exactly it optimizes. The second covers the existing constructions, how far they reach, where the first unknown orders lie, and what is proved about the problem as a whole. The third describes an application in numerical rounding. It shows that the perfect balance the conjecture promises does not guarantee less rounding error. Readers interested only in the conjecture can stop after the second part.

Part I. The object

Why multiples of four appear

An order-n Hadamard matrix is a real matrix H with entries in $\{-1, +1\}$ satisfying

$$HH^{\mathsf{T}} = nI_n .$$

Each row has squared length n, and distinct rows have dot product zero. Consequently $Q = H/\sqrt{n}$ is orthogonal, so $H^{\mathsf{T}} H = nI_n$ as well and the columns are orthogonal too. The normalization matters: H itself multiplies lengths by $\sqrt{n}$.

Two operations preserve the defining identity: permuting rows or columns, and multiplying a row or column by −1. Matrices related by these operations are called *equivalent*. Every Hadamard matrix is equivalent to a *normalized* one, whose first row and first column contain only +1.

Orders one and two have immediate examples, (1) and $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$. For $n > 2$, orthogonality forces n to be divisible by four. Here is the complete counting argument.

Flip whole columns until the first row consists entirely of +1. Column flips preserve row dot products. The second row must then contain equally many +1 and −1 entries, so n is even. Reorder the columns so the second row has its positive entries in the first half.

Let the third row have a positive entries in the first half and b in the second. Orthogonality with the first row gives $a + b = n/2$. Orthogonality with the second gives $a = b$. Therefore $a = b = n/4$ must be an integer.

This is a necessary condition, not a construction. **The Hadamard conjecture asserts the converse: for every positive integer k, there is a Hadamard matrix of order $4k$.** Confusing these directions would turn a short counting proof into a false solution.

An exact optimization meaning

For a real orthogonal matrix Q, define its largest entry magnitude

$$\mu(Q) = \max_{i,j} |Q_{ij}|.$$

Every row has squared length one. If all entries were smaller in magnitude than $1/\sqrt{n}$, the sum of their squares would be smaller than one. Thus

$$\mu(Q) \geq \frac{1}{\sqrt{n}}.$$

Equality forces every entry of every row to have magnitude exactly $1/\sqrt{n}$. Otherwise a row containing a smaller entry could not reach squared length one without another entry exceeding the bound. Hence $\sqrt{n}\,Q$ is a Hadamard matrix. Conversely, $H/\sqrt{n}$ attains equality for every Hadamard matrix H.

This is more than a statement about individual entries. Treat x as a row vector. For $\| x \|_1 \leq 1$,

$$|(xQ)_j| \leq \sum_i |x_i|\,|Q_{ij}| \leq \mu(Q).$$

Choose x to be a coordinate vector corresponding to an entry attaining the maximum, and equality is possible. Therefore

$$\sup_{\|x\|_1 \le 1} \|xQ\|_\infty = \mu(Q).$$

The worst-case peak is therefore always at least $1/\sqrt{n}$, **and this lower bound is attained exactly when an order-n Hadamard matrix exists**. A minimum over all orthogonal matrices always exists, because the real orthogonal group is compact and the largest-entry function is continuous. When no Hadamard matrix of order n exists, that minimum is strictly larger than $1/\sqrt{n}$.

The Hadamard conjecture is thus an exact optimality question: does this lower bound become attainable in every dimension $n = 4k$? The connection with flat real orthogonal matrices is established mathematics. Jaming and Matolcsi study this viewpoint and relaxations of perfect flatness. [2]

The input constraint is essential. Replace the ℓ_1 unit ball by the ℓ_2 unit sphere and every orthogonal matrix has the same worst-case peak:

$$\sup_{\|x\|_2 = 1} \|xQ\|_\infty = 1.$$

The upper bound follows from length preservation. The vector $x = e_j Q^{\mathsf{T}}$ attains it. No fixed orthogonal transformation makes every unit vector less spiky.

The same object maximizes a determinant

Hadamard's own interest was a determinant problem. He proved the bound below for complex entries of modulus at most one, and showed that real matrices attaining it need order divisible by four. [3] For any real $n \times n$ matrix A with entries of magnitude at most one, let $G = AA^{\mathsf{T}}$. Its nonnegative eigenvalues have sum $\operatorname{tr} G \le n^2$. Applying the arithmetic-geometric mean inequality gives

$$\det G \le \left(\frac{n^2}{n}\right)^n = n^n,$$

and hence $|\det A| \le n^{n/2}$. For a sign matrix, equality requires all eigenvalues of G to equal n. Since G is real symmetric, this is equivalent to $G = nI$.

So a Hadamard matrix simultaneously saturates a determinant bound and, after normalization, the smallest possible largest-entry bound. These are two precise meanings of perfect balance. When n is not a multiple of four, the largest determinant of a sign matrix is a separate and harder *maximal determinant problem*. It is open in general. [4]

Equivalent combinatorial forms

The conjecture has several equivalent forms, which is one reason it appears in so many fields. [4,5]

- **Designs.** Normalize an order-$4t$ Hadamard matrix and delete its first row and column. Record each remaining +1 as an incidence. The result is a symmetric 2-$(4t-1,\ 2t-1,\ t-1)$ design, and every such design arises in this way. The conjecture is equivalent to the existence of these *Hadamard designs* for every t.
- **Codes.** The rows of H and $-H$, written over $\{0, 1\}$, form a binary code of length n with $2n$ codewords and minimum distance $n/2$. Plotkin's bound shows this is the largest possible such code, so Hadamard matrices give optimal codes.
- **Orthogonal arrays and experimental design.** Hadamard matrices give two-level orthogonal arrays of strength two. In statistics these are the Plackett-Burman designs for screening many factors with few runs.

These are reformulations, not routes around the difficulty. Being equivalent, none is easier than the original.

Part II. What is known

Building matrices: products and doubling

The simplest construction combines two smaller matrices. If H_m and H_n are Hadamard, their Kronecker product $H_m \otimes H_n$ is Hadamard of order mn, because

$$(H_m \otimes H_n)(H_m \otimes H_n)^{\mathsf{T}} = H_m H_m^{\mathsf{T}} \otimes H_n H_n^{\mathsf{T}} = mn\, I_{mn}.$$

Taking H_2 as one factor gives Sylvester's 1867 doubling: [6]

$$H_{2n} = \begin{pmatrix} H_n & H_n \\ H_n & -H_n \end{pmatrix}.$$

Starting at order one gives every power of two. It does not give order twelve. Infinite is not the same as exhaustive. Products alone also never produce an order whose odd part is a new prime, so new building blocks are needed. Hadamard himself exhibited matrices of orders 12 and 20 in 1893. [3] In 1898 Scarpis showed that a Hadamard matrix of order $p + 1$, for a prime $p \equiv 3 \pmod 4$, yields one of order $p(p + 1)$. [4]

Paley's construction and order twelve

Paley's 1933 paper supplied the first systematic infinite families of new orders, from finite fields. [7] It also states the conjecture: "It seems probable that, whenever m is divisible by 4, it is possible to construct an orthogonal matrix of order m composed of ±1, but the general theorem has every appearance of difficulty." [7, p. 312] We prove the prime-field case of his first construction.

Take a prime $q \equiv 3 \pmod 4$. Define the quadratic character χ on residues modulo q: it is zero at zero, +1 at nonzero squares, and −1 at non-squares. Let

$$C_{ij} = \chi(j - i), \qquad 0 \le i, j < q.$$

The nonzero squares form a subgroup of index two, so χ is multiplicative and exactly half the nonzero residues have each sign. Every row sum of C is therefore zero. Also -1 is not a square: a solution of $t^2 = -1$ would give $t^{q-1} = -1$ when $q \equiv 3 \pmod 4$, contrary to Fermat's little theorem. Thus $\chi(-1) = -1$, giving $C^{\mathsf{T}} = -C$.

The needed correlation identity is

$$\sum_{x} \chi(x - a)\chi(x - b) = \begin{cases} q - 1, & a = b, \\ -1, & a \ne b. \end{cases}$$

For distinct a, b, set $u = (2x - a - b)/(b - a)$. Then $(x - a)(x - b) = ((b - a)/2)^2 (u^2 - 1)$. The nonzero square factor has character one, so this invertible change of variable reduces the sum to $\sum_u \chi(u^2 - 1)$. Count pairs satisfying $v^2 = u^2 - 1$. For each u there are $1 + \chi(u^2 - 1)$ choices of v. On the other hand, $(u - v)(u + v) = 1$. Each nonzero residue $t = u - v$ uniquely determines $u + v = t^{-1}$, then u and v because two is invertible. There are $q - 1$ pairs. Thus $q + \sum_u \chi(u^2 - 1) = q - 1$, proving the identity.

It follows that $CC^{\mathsf{T}} = qI - J$, where J is the all-ones matrix. Now set

$$H = \begin{pmatrix} 1 & \mathbf{1}^{\mathsf{T}} \\ \mathbf{1} & C - I \end{pmatrix}.$$

Every entry is a sign. The first row is orthogonal to every later row because $C - I$ has row sum -1. The lower-right block of HH^{T} is

$$J + (C - I)(C - I)^{\mathsf{T}} = J + (qI - J) + I = (q + 1)I.$$

Therefore $HH^{\mathsf{T}} = (q + 1)I$. The same argument works over any finite field of order $q \equiv 3 \pmod 4$, so Paley's first construction gives order $q + 1$ for every such prime power. His second construction gives order $2(q + 1)$ for every prime power $q \equiv 1 \pmod 4$. [4,7]

For $q = 11$, the nonzero squares are $1, 3, 4, 5, 9$. The resulting order-twelve matrix is below. A plus or minus denotes an integer, not an approximation.

```
+ + + + + + + + + + + +
+ - + - + + + - - - + -
+ - - + - + + + - - - +
+ + - - + - + + + - - -
+ - + - - + - + + + - -
+ - - + - - + - + + + -
+ - - - + - - + - + + +
+ + - - - + - - + - + +
+ + + - - - + - - + - +
+ + + + - - - + - - + -
+ - + + + - - - + - - +
+ + - + + + - - - + - -
```

The exact verifier, `submission/v2/strengthening/exact_checks.py`, checks all 144 row dot products and computes $|\det H| = 12^6 = 2,985,984$ using fraction-free integer elimination. It also checks the construction at primes $3, 7, 11, 19, 23, 31, 43$, producing orders $4, 8, 12, 20, 24, 32, 44$.

How far the classical methods reach

Combine Sylvester doubling, both Paley constructions over prime powers, and Kronecker products. The accompanying script `known_orders.py` computes exactly which orders these classical methods reach. Up to 200 they give 44 of the 50 multiples of four. The six missing orders are

$$92,\ 116,\ 156,\ 172,\ 184,\ 188.$$

These are exactly the orders below 200 that Paley himself listed as unsettled. Order 92 is the first. Paley's first construction would need $q = 91 = 7 \cdot 13$, and his second would need $q = 45 = 3^2 \cdot 5$. Neither is a prime power. The only factorizations of 92 into two orders greater than one use 2 and 46, or 4 and 23, and neither 46 nor 23 is a Hadamard order. Williamson's 1944 paper gave a matrix of order 172, order 92 fell in 1962, and $184 = 2 \cdot 92$ then followed by doubling. [7,8,9]

Up to 1000, these three methods alone reach 195 of the 250 multiples of four. Product variants, such as Agaian's theorem giving order $8mn$ from orders $4m$ and $4n$, reach a few more. [4] A missed order is not an unknown order. The classical methods miss many orders below 2000, and every one of those has since been reached by other methods.

Williamson's method and the computer

Williamson's 1944 method builds an order-$4m$ matrix from four smaller pieces placed in a fixed array. [8] The following is one standard form. Let A, B, C, D be symmetric circulant $m \times m$ sign matrices satisfying

$$A^2 + B^2 + C^2 + D^2 = 4m I_m .$$

Then the block matrix

$$H = \begin{pmatrix} A & B & C & D \\ -B & A & -D & C \\ -C & D & A & -B \\ -D & -C & B & A \end{pmatrix}$$

is Hadamard of order $4m$. Circulant matrices always commute with each other, and symmetry makes each block equal to its transpose. The pattern of signs is the multiplication table of the quaternions, which is why the cross terms cancel. For $m = 3$, take $A = J$ and $B = C = D = 2I - J$. Then $A^2 + 3B^2 = 3J + 3(4I - 4J + 3J) = 12I$, giving another order-twelve matrix. The accompanying test checks this block identity directly.

The method turns a search over $(4m)^2$ signs into a search over four circulant first rows of length m, with symmetry cutting that further. In 1962 Baumert, Golomb and Hall used an early computer at the Jet Propulsion Laboratory to find Williamson matrices with $m = 23$. This gave the first Hadamard matrix of order 92. [9] It is an early example of a computer settling a case of a classical existence question.

It was once conjectured that every order $4m$ has a Hadamard matrix of Williamson type. That *Williamson conjecture* is false. Exhaustive searches show that Williamson matrices do not exist for $m = 35$, nor for $m = 47, 53, 59$. [10] Hadamard matrices of those orders exist by other means. Turyn found an infinite family: Williamson matrices of size $(q + 1)/2$ for every prime power $q \equiv 1 \pmod 4$. [12] Goethals and Seidel removed the symmetry requirement by inserting the reversal matrix, with ones on the back diagonal, into a modified array. They introduced it in 1970 to build skew-Hadamard matrices; its general form takes any four circulant sign matrices whose Gram matrices sum to $4ml$. [11] T-sequences, built from base sequences or Turyn-type sequences, can be placed in the Goethals-Seidel array or in larger Baumert-Hall arrays, and reach many more orders. Later work added difference families, further infinite families, and product theorems that build new orders from old ones. [4] Each method reaches many new orders, and none is known to reach them all.

The computational frontier

Progress at individual orders came from large structured searches. Baumert found order 116 in 1966. By 1977 the smallest open order was 268; Sawade settled it in 1985, which left 428. [28] Kharaghani and Tayfeh-Rezaie found order 428 in 2004 by a cluster search for Turyn-type sequences. These gave T-sequences, which they placed in a Goethals-Seidel array; the paper appeared in 2005. [13] Đoković constructed order 764 the same way, posted in 2007 and published in 2008. [14]

For about twenty years after 428, the smallest open order was $668 = 4 \cdot 167$. Paley's constructions would need $q = 667 = 23 \cdot 29$ or $q = 333 = 3^2 \cdot 37$, and neither is a prime power. Products do not help either: the only factorizations of 668 into two orders greater than one use 2 and 334, or 4 and 167, and neither 334 nor 167 is an admissible order. In August 2026 Alpöge, Voinov and Reynolds-Haertle announced Hadamard matrices at the twelve orders below 2000 at which none had been known:

668, 716, 892, 1132, 1244, 1388, 1436, 1676, 1772, 1916, 1948, 1964.

The announcement gave the matrices but not how they were found. Epoch AI's open-problem page provisionally records the result as obtained by a team of three people working with an AI system. [29] In October 2026 Egan, Kotsireas, London and Ó Catháin verified the twelve matrices and analysed their structure. Each is a Goethals-Seidel array over an abelian group, nine over cyclic groups and three over non-cyclic groups, some with a border of extra rows and columns. [28] A Hadamard matrix is therefore known at every multiple of four below 2000.

The same survey lists the 95 multiples of four up to 10,000 at which no Hadamard matrix is currently known. The smallest is now $2396 = 4 \cdot 599$. [28] Paley's constructions would need $q = 2395 = 5 \cdot 479$ or $q = 1197 = 3^2 \cdot 7 \cdot 19$, and the only factorizations into two orders greater than one, $2 \cdot 1198$ and $4 \cdot 599$, again use non-admissible factors. A missing matrix is evidence of difficulty, not of nonexistence. No order divisible by four is known to be impossible.

The number of matrices grows quickly once they exist. Up to equivalence there is one Hadamard matrix of each order 1, 2, 4, 8 and 12. Order 16 has five, order 20 has three, order 24 has 60 and order 28 has 487. [16] Order 32 has exactly 13,710,027, a classification Kharaghani and Tayfeh-Rezaie completed in 2013. [17] Abundance at the orders we can search does not translate into a general existence proof.

What is proved about all orders

The strongest general results show that every odd number appears in some Hadamard order, provided we allow a large enough power of two. In 1976 Seberry proved that for every odd q there is a Hadamard matrix of order $2^t q$ for all sufficiently large t. [18] Her paper stated the explicit threshold $t \approx 2 \log_2 (q - 3)$. A 2007 note by Jensen showed that the proof gives a threshold one larger than stated, already for primes $p \equiv 1 \pmod 4$, so the stated bound does not follow and its extension to composite q breaks down. The existence of some threshold is unaffected. [18] Craigen used *signed groups* to sharpen this asymptotic estimate, lowering the coefficient of $\log_2 q$ to 2/3. [19] Craigen and Tiessen reduced it to 3/8 in unpublished work, and Livinskyi's 2012 thesis to 1/5. [20] A 2024 preprint argues that the 1/5 step relies on an incorrect lemma, accepts Livinskyi's weaker coefficient 3/13, and claims 3/20 itself; it had not been published at the time of writing. [20] The conjecture asks for $t = 2$ for every odd q, and all of these bounds still grow with q.

Density results measure the same gap from another direction. If the conjecture holds, the number of Hadamard orders up to x is about $x/4$. Paley's constructions alone give on the order of $x / \log x$ orders. Counting Kronecker products of Paley orders, de Launey and Gordon proved the slightly larger lower bound $x\, e^{(C + o(1))(\log \log \log x)^2} / \log x$, with $C \approx 0.8178$. They also showed that the other major constructions do not improve this bound. It is not even known that the Hadamard orders have positive density. [21] De Launey proved that for every $\varepsilon > 0$, a positive proportion of odd numbers k admit a Hadamard matrix of order $k \cdot 2^{2 + \lfloor \varepsilon \log_2 k \rfloor}$. [22] Allowing only a slowly growing power of two thus covers a positive fraction of odd parts, but not all of them.

Two related problems are also open and are sometimes confused with this one. The *skew-Hadamard conjecture* asks for a Hadamard matrix with $H + H^{\mathsf{T}} = 2I$ in every order divisible by four. It is stronger than the Hadamard conjecture and has its own frontier: as of 2025, Cati and Pasechnik list 41 orders up to 1208 with no known skew-Hadamard matrix. [15] Ryser's *circulant Hadamard conjecture* asserts the opposite kind of statement: no circulant Hadamard matrix exists beyond order four. Excluding circulant matrices does not construct general Hadamard matrices. [1,4]

Why the problem is hard

Three features explain why no single idea has closed the gap.

1. **Constructions need structure.** Every infinite family comes from algebra: finite fields, group rings, or sequences with special autocorrelation. Each structure exists only at orders with matching arithmetic, such as $q + 1$ for a prime power q. The open orders are exactly those where no known structure fits.
2. **Unstructured search is hopeless at these sizes.** An order-2396 sign matrix has $2396^2 = 5,740,816$ entries. Williamson-type arrays reduce this to a few sequences of length 599. Even that space is too large to search exhaustively without further constraints that might exclude every solution.
3. **Naive probability gives the wrong answer.** For a random sign matrix, two given rows are orthogonal with probability about $\sqrt{2/(\pi n)}$. If the $\binom{n}{2}$ pairs were independent, the expected number of Hadamard matrices would tend to zero as n grows. Yet order 32 alone has more than thirteen million inequivalent examples. The orthogonality events are strongly dependent, so this naive reasoning cannot settle existence either way. Rigorous probabilistic methods can count matrices with a small number of mutually orthogonal rows. [23] They are nowhere near the full square case.

The conjecture is therefore not about whether balance is plausible. It asks whether a rigid algebraic coincidence, which we can produce in many families, occurs in every admissible order.

Part III. An application: balance is not rounding

The rotation trick

Large numerical models are often stored with low-precision weights and activations. A single large coordinate forces a coarse quantization grid on everything that shares its scale. A common remedy rotates the coordinates first. For a linear layer $Y = XW$ and any orthogonal Q,

$$X_r = XQ, \qquad W_r = Q^{\mathsf{T}} W, \qquad X_r W_r = XW.$$

A scaled Hadamard matrix $H/\sqrt{n}$ is the natural choice. By Part I it minimizes the worst-case peak over the ℓ_1 ball, and a Sylvester matrix also has a fast transform. Methods such as QuaRot and QuIP# build randomized Hadamard rotations into language-model quantization. SpinQuant shows that the choice of rotation matters. [24,25,26] This article does not propose or evaluate such a method.

The question here is narrower. Suppose the conjecture were true, so perfectly balanced rotations existed in every admissible dimension. Would they guarantee lower rounding error? They would not. This holds even with the optimal peak, and even for every Hadamard matrix at once.

A fixed input that every Hadamard matrix hurts

Use a symmetric max-absolute-value quantizer. At bit width b, let $K = 2^{b-1} - 1$. For a nonzero vector z, its scale is $\Delta = \|z\|_\infty / K$, and its reconstructed coordinates are $\hat{z}_i = \Delta \operatorname{round}(z_i/\Delta)$, with nearest rounding, ties to even, and signed codes $-K, \dots, K$. A smaller peak gives a finer grid, but the distance from the data to the grid also matters.

Theorem. Fix an order $n \geq 4$ for which a real Hadamard matrix exists, a bit width $b \geq 3$, and three distinct indices a, c, d. Let

$$x = Ke_a + e_c + e_d, \qquad w = e_a.$$

For **every** order-n Hadamard matrix H, put $Q = H/\sqrt{n}$ and quantize each vector with one full-width scale. Then x and w are represented exactly, and the rotated peak $(K+2)/\sqrt{n}$ is smaller than K. But the rotated activation and the rotated linear output both have positive error. Their relative squared values depend on neither n nor H:

Bit width	Activation reconstruction error	Linear-output error
$b = 3$	1/66	1/81
$b = 4$	11/3332	1/784
$b \geq 5$	$24/[K^2(K^2+2)]$	$16/K^4$

Proof. The original scale is one, so x and w are exact. Define sign sequences $p_j = H_{aj}H_{cj}$ and $q_j = H_{aj}H_{dj}$. Orthogonality of rows a, c and d gives

$$\sum_j p_j = 0, \qquad \sum_j q_j = 0, \qquad \sum_j p_j q_j = \sum_j H_{cj}H_{dj} = 0.$$

For $s, t \in \{-1, +1\}$, the number of columns with $p_j = s$ and $q_j = t$ is therefore

$$N_{s,t} = \frac{1}{4}\sum_j (1 + sp_j)(1 + tq_j) = \frac{n}{4}.$$

This is the same three-row count that forced n to be a multiple of four. Write $u = xH$ and $v_j = H_{aj}u_j = K + p_j + q_j$. Thus v_j takes the values $K+2, K, K-2$ with multiplicities $n/4, n/2, n/4$, and the rotated peak is $(K+2)/\sqrt{n} < K$.

Work before dividing by $\sqrt{n}$, with scale $\Delta = (K+2)/K$. Symmetric rounding is odd, so the sign H_{aj} passes through it. For $b \geq 5$, so $K \geq 15$, the codes for $K+2, K, K-2$ are $K, K-2, K-4$, because

$$\frac{K^2}{K+2} = K - 2 + \frac{4}{K+2}, \qquad \frac{K(K-2)}{K+2} = K - 4 + \frac{8}{K+2},$$

and both fractions are below one half. The reconstruction errors are $0,\ -4/K,\ -8/K$. Their squared values, averaged with the multiplicities above, give $24/K^2$. Dividing by $\|x\|_2^2 = K^2 + 2$ gives the activation formula.

The rotated weight $Q^{\mathsf{T}}w$ has every coordinate equal to $\pm 1/\sqrt{n}$, so its quantizer is exact. The quantized output is the average of the rounded v_j, which is $K - 4/K$, whereas the exact output is K. The relative squared error is $16/K^4$. At three bits, scale 5/3 rounds $5, 3, 1$ to $5, 10/3, 5/3$. At four bits, scale 9/7 rounds $9, 7, 5$ to $9, 45/7, 36/7$. Direct substitution gives the remaining table entries. ▫

As the dimension grows, the peak ratio $(K + 2)/(K\sqrt{n})$ tends to zero while the error stays fixed. Choosing a different Hadamard matrix, or randomizing its signs and permutations, cannot remove the error for this input, because those operations produce another Hadamard matrix. This does not contradict probabilistic guarantees for random inputs. It shows that perfect balance alone does not guarantee a better result. The theorem assumes one scale per vector. It says nothing about smaller groups, learned quantizers or real models. No first-discovery claim is made for this derivation.

The accompanying checker evaluates 1,280 rational cases across seven Paley orders and one Sylvester order, including the non-power-of-two orders 12, 20, 24 and 44. A further 672 cases cover Sylvester matrices with every supported sign pattern. Finite cases corroborate the proof; they do not replace it.

Test the counterexample family

Change the dimension, bit width or signs. The peak falls, but rounding introduces error where the original input had none. This calculator evaluates every transformed coordinate with exact integer arithmetic. Only the peak display uses a floating-point square root.

Peak magnitude

7 → 4.500000

64.286% of the original peak

Relative squared error after rotation

0.3301321%

Exact fraction: 11/3332

Original-basis reconstruction error is exactly zero. At a fixed bit width, changing the dimension or signs leaves the positive error fraction unchanged. Interactive controls are available in the HTML edition.

A fresh-seed study with competing transforms

A constructed example shows what can happen; it does not show what usually happens. A separate study compares four transforms on synthetic data. Its protocol fixed seeds 1000 through 1099, every setting and one primary comparison before any of those samples were generated. [27]

Each seed creates a 512×128 Gaussian activation matrix and a 128×64 Gaussian weight matrix. Activation column zero is multiplied by $1, 2, 4, 8, 16$ or 32 to create an outlier. Four arms are compared: the original basis, a randomized Hadamard rotation, a Haar-random orthogonal matrix, and a signed permutation that reorders without mixing. Both operands receive the same four-bit quantizer, grouped over 32 input features in the primary setting. Error is the relative squared Frobenius error of the output, simulated in float64.

At the primary setting, with multiplier 32, mean error fell from **2.606175% to 1.355898%** under Hadamard, a **47.97% reduction**. All 100 paired seeds favored Hadamard, and the 95% whole-seed bootstrap interval for the change was [− 1.307, − 1.194] percentage points.

The secondary comparisons limit what this shows. The table gives mean percentage errors at group size 32 for selected multipliers.

Activation-column multiplier	Original	Hadamard	Haar orthogonal	Signed permutation
1	1.891841	1.899478	1.895267	1.892540
4	2.237005	1.878227	1.892689	2.229054
16	3.725695	1.572253	1.887476	3.705417
32	2.606175	1.355898	1.893716	2.577981

- **Generic rotations also help.** A Haar-random orthogonal matrix also removes most of the outlier penalty, so the benefit is not exclusive to Hadamard.
- **No outlier, no benefit.** With no outlier, Hadamard was marginally worse. For isotropic Gaussian inputs every orthogonal rotation leaves the error distribution unchanged, so this small difference is sampling noise, not a systematic effect.
- **Grid-aligned data is hurt.** When the data already sit on the quantization grid, the original basis has zero error. Hadamard averaged 2.742108% error and lost on all 100 seeds. This is the mechanism of the theorem above, seen in random data.

The study retains all 7,600 measurements, input and transform hashes, and an independent audit that recomputes every reported summary from the raw data. It measures no model quality, memory or speed.

What the evidence supports

The Hadamard conjecture asks whether a perfectly balanced orthogonal matrix exists in every dimension divisible by four. Part I showed that this balance is exactly the minimum possible largest entry and the maximum possible determinant. Part II showed how far algebraic and computer-assisted constructions have reached: every order below 2000, and every odd factor once enough powers of two are allowed. The general case remains open. Part III showed that the promised balance does not guarantee less rounding error. Every Hadamard matrix fails on the same fixed input, while a synthetic study shows the benefit in the case the method targets.

The constructions, equivalences and frontier results in Parts I and II are established mathematics, credited to their sources. Proofs are given where they are short enough to include. The counterexample theorem and the study are offered as teaching material, not as new quantization methods. Nothing here proves the conjecture at any new order.

Reproduce and inspect

The checks need Python and NumPy, with no GPU, model download or external dataset. Run them from the repository root.

```
python submission/v3/known_orders.py --limit 200
python -m unittest discover -s submission/v3 -p 'test_*.py' -v
cd submission/v2
python -m unittest discover -s strengthening -p 'test_*.py' -v
python strengthening/audit_study.py run-v2
python strengthening/all_hadamard_checks.py > new-all-hadamard-checks.json
```

The first two commands check the classical-order counts and the Williamson examples in Part II. The remaining commands check the Paley construction, the counterexample theorem and the retained study data. Those are unchanged from version 2 and stored in `submission/v2`. A fresh study run uses `python strengthening/study.py --output new-run` from that directory.

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